MagCNN — Physics-Informed Attention U-Net Results

ECE 228 Final Project | TensorFlow 2.20 | Colab T4 GPU | Two Training Runs

1. Model Architecture & Configuration

MagCNN is a 16M-parameter attention U-Net built in TensorFlow/Keras. It takes 6-channel XMCD scattering amplitude maps as input (one azimuthally-averaged channel per polar angle) and outputs the 3-component spin vector field M(x,y) = (Mx, My, Mz) at each pixel. The architecture uses residual blocks with GELU activations, batch normalization, and attention gates in the skip connections.

Parameter	Value	Parameter	Value
Architecture	Attention U-Net	Parameters	16,288,375
Framework	TensorFlow 2.20.0	Hardware	Colab T4 GPU
Grid size	64 x 64	Input channels	6 (one per polar angle)
Output channels	3 (Mx, My, Mz)	Training samples	4,000
Val samples	500	Epochs planned	60
Batch size	32	Optimizer	Adam
Activation	GELU	Skip connections	Attention gates
Data gen rate	~19 samples/s	Data gen time	250.7 s (~4.2 min)

2. Forward Scattering Model (Born Approximation)

Parameter	Value
Model	Single-layer Born approximation: A(q) = FFT2[k-hat · M] x XMCD asymmetry
Polar angles (theta)	6 values
Azimuthal samples (phi)	36 per polar angle (216 total projections per sample)
Input to network	6 channels — azimuthally averaged |A(q)| per theta angle
Physics encoded	Incident beam direction k-hat, XMCD contrast, FFT projection

3. Loss Function Design — Two Versions

Version	Loss	Rationale
v1 (baseline)	Pure MSE: L = mean((Y_true - Y_pred)^2)	Standard pixel-wise reconstruction loss. Simple baseline.
v2 (physics)	Combined: MSE + w_norm*||1-|M_pred|||^2 + w_tv*TV(M_pred) + epoch-delayed angular term	Unit norm constraint forces |M|=1. TV regularizes smooth domain walls. Angular term added after MSE warmup.

4. Training Run v1 — Pure MSE Loss (31/60 epochs completed)

Epoch	Train Loss	Train Ang	Val Ang	Notes
1	1.9924	87.76°	87.68°	Near-random at start
10	1.9581	87.38°	87.79°	Flatlined — no improvement
15	1.9577	87.37°	90.39°	Val spikes above random (90°)
16	1.9580	87.38°	91.05°	Val exceeds random baseline
20	1.9579	87.38°	92.00°	Val at 92° — WORSE than random
22	1.9571	87.34°	87.67°	Briefly recovers
27	1.9571	87.34°	91.37°	Spikes again
31	1.9578	87.37°	90.53°	Run stopped

Metric	Value
Train loss change over 31 epochs	1.9924 → 1.9571 (delta = 0.035 — essentially zero)

Train angular error range	87.76° → 87.34° (delta = 0.42° — flatlined)
Val angular error	Oscillating 87°–92°, peaked at 92.00° (worse than random)
Random baseline	~90°
Assessment	Complete failure. Model collapsed to predicting near-zero vectors everywhere.

FAILURE MODE v1: Classic mean-prediction collapse. The model learned to output near-zero vectors everywhere, minimizing MSE but encoding no directional information. Val angular error oscillated around and above the 90° random baseline. After 31 epochs the loss had dropped only 0.035.

5. Training Run v2 — Physics-Informed Loss (14/60 epochs completed)

Epoch	Train Loss	Train Ang	Val Ang	Notes
1	0.4799	88.07°	87.69°	Loss scale lower — norm constraint active
2	0.5374	87.66°	87.67°	Loss INCREASING — norm term ramping up
5	0.7688	87.75°	87.98°	Loss still climbing
10	1.1478	87.46°	87.88°	Loss >1.0 and still rising
14	1.4525	87.46°	87.66°	Run stopped — loss climbing, no ang improvement

Metric	Value
Train loss trajectory	0.4799 → 1.4525 (monotonically INCREASING over 14 epochs)
Train angular error range	88.07° → 87.46° (delta = 0.61° — flat)
Val angular error	87.69° → 87.66° (completely flat)
Assessment	Physics regularization terms dominate and destabilize training. Loss diverges.

FAILURE MODE v2: Combined loss increased monotonically 0.48 → 1.45 over 14 epochs. The norm + TV regularization weights were too large relative to MSE, causing the optimizer to chase regularization objectives rather than learning the inverse map. Angular error remained flat at ~87.5° throughout — identical to v1. Run stopped at epoch 14.

6. v1 vs v2 Head-to-Head

Metric	v1 (Pure MSE)	v2 (Physics-Informed)	Verdict
Loss trajectory	Slowly decreasing	Monotonically increasing	v1
Loss delta over run	0.035 over 31 ep	+0.972 over 14 ep	Neither converged
Train ang final	87.34° (ep 31)	87.46° (ep 14)	Tie — both flat
Val ang stability	Oscillating 87–92°	Flat ~87.7°	v2 more stable
Learning signal	Minimal	None	Neither
Overall	Flat/oscillating failure	Diverging loss failure	Both fail

7. Root Cause Analysis

#	Cause	Evidence
1	Phase retrieval — amplitude-only input	XMCD measures |A(q)|^2, not A(q). Phase lost. Inverse map from amplitude to spin field is ill-posed. MSE loss cannot provide orientation gradient from magnitude data alone.
2	Azimuthal averaging destroys directional info	6 input channels are azimuthally averaged — collapsing 2D Fourier pattern to 1D radial profile. All information about which in-plane direction the spin points is discarded. Mx and My are fundamentally unrecoverable from this input.
3	Mean-prediction collapse under MSE	MSE minimized by predicting the mean of the training distribution (~zero for random domains). Model converges to predict zero everywhere, giving ~87° angular error.
4	Physics loss miscalibration in v2	Norm constraint and TV weighted too strongly. Combined loss increased rather than decreased. Optimizer chased regularization objectives rather than reconstruction.

5	Bottleneck collapse (earlier versions)	4-level encoder reached 4x4 bottleneck at 64x64 input, destroying spatial structure. Fixed in v2 by removing deepest encoder level (bottleneck now 8x8).

8. Significance for Paper

Finding	Implication
Both MagCNN runs fail — angular error stuck at ~87°	Even 16M-param attention U-Net with physics loss cannot solve 2D XMCD inverse without phase info. Motivates need for explicit phase retrieval or richer multi-angle input representations.
Classical LS direct achieves 55° angular error vs 87° for ML	The direct linear solver on the MATLAB 3D pipeline outperforms the ML approach on angular error despite being far simpler. Ill-posedness is the bottleneck, not model capacity.
Benchmark (5-model run) shows U-Net learns Mz but not Mx/My	Single-angle amplitude inputs cannot disambiguate in-plane spin components. Multi-angle conditioning partially helps Mz but in-plane collapse persists across all architectures.
Physics-informed losses require careful weight scheduling	Naive addition of norm + TV constraints without warmup actively harms convergence. Weight tuning is a prerequisite, not an afterthought.